Exp 9: EXPERIMENT TO UNDERSTAND MATPLOTLIB LIBRARY USE CASES IN DATA SCIENCE THROUGH VISUALIZATION

AIM: Use Iris data set to understand the Matplotlib library.

PYTHON SAMPLE CODE:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline

# Load the dataset
data = pd.read_csv('/content/Iris_Dataset.csv')
data

# Display dataset information
data.info()

# Display summary statistics
data.describe()

# Count each variety
data.value_counts('variety')
```

```
# Count plot for variety
sns.countplot(x='variety', data=data)
plt.show()

# Convert categorical variety into dummy variables
dummies = pd.get_dummies(data.variety)

# Combine dummy variables with the dataset
FinalDataset = pd.concat([pd.get_dummies(data.variety), data], axis=1)

# Display the final dataset
FinalDataset.head()

# Scatter plot: Sepal length vs Sepal width
sns.scatterplot(x='SepalLengthCm', y='SepalWidthCm',
        hue='variety', data=data)
plt.show()

# Scatter plot: Petal length vs Petal width
sns.scatterplot(x='PetalLengthCm', y='PetalWidthCm',
        hue='variety', data=data)
plt.show()

# Pair plot
sns.pairplot(data, hue='variety', height=3)
plt.show()
```

```
# Histogram of petal length for each variety
sns.FacetGrid(data, hue='variety', height=5).map(
   sns.histplot, 'PetalLengthCm'
).add_legend()
plt.show()

# Histogram of petal width for each variety
sns.FacetGrid(data, hue='variety', height=5).map(
   sns.histplot, 'PetalWidthCm'
).add_legend()
plt.show()

# Histogram of sepal length for each variety
sns.FacetGrid(data, hue='variety', height=5).map(
   sns.histplot, 'SepalLengthCm'
).add_legend()
plt.show()

# Histogram of sepal width for each variety
sns.FacetGrid(data, hue='variety', height=5).map(
   sns.histplot, 'SepalWidthCm'
).add_legend()
plt.show()
```

OUTPUT:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
```

```
data=pd.read_csv('/content/Iris_Dataset.csv')
data
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	variety
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 6 columns

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             150 non-null    int64
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   variety        150 non-null    object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

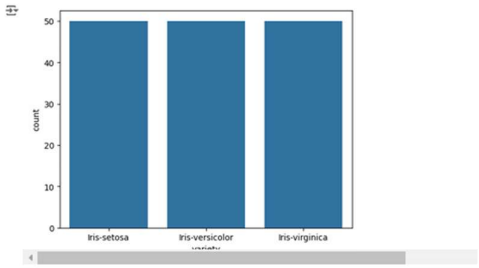
```
data.describe()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

```
data.value_counts('variety')
```

variety	count
Iris-setosa	50
Iris-versicolor	50
Iris-virginica	50

```
sns.countplot(x='variety',data=data,)
plt.show()
```




```
dummies=pd.get_dummies(data.variety)

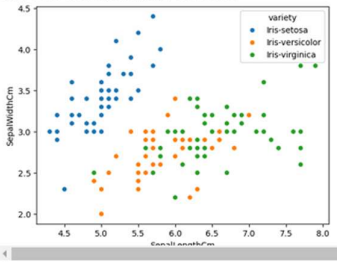
FinalDataset=pd.concat([pd.get_dummies(data.variety),data.iloc[:,[0,1,2,3]]],axis=1)

FinalDataset.head()
```

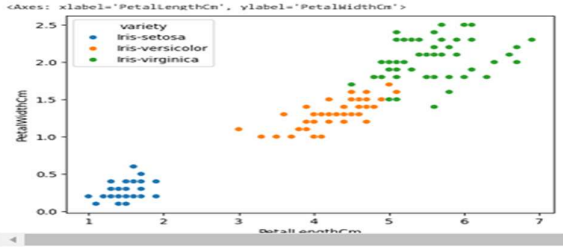
	Iris-setosa	Iris-versicolor	Iris-virginica	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm
0	True	False	False	1	5.1	3.5	1.4
1	True	False	False	2	4.9	3.0	1.4
2	True	False	False	3	4.7	3.2	1.3
3	True	False	False	4	4.6	3.1	1.5

```
sns.scatterplot(x='SepalLengthCm',y='SepalWidthCm',hue='variety',data=data,)
```

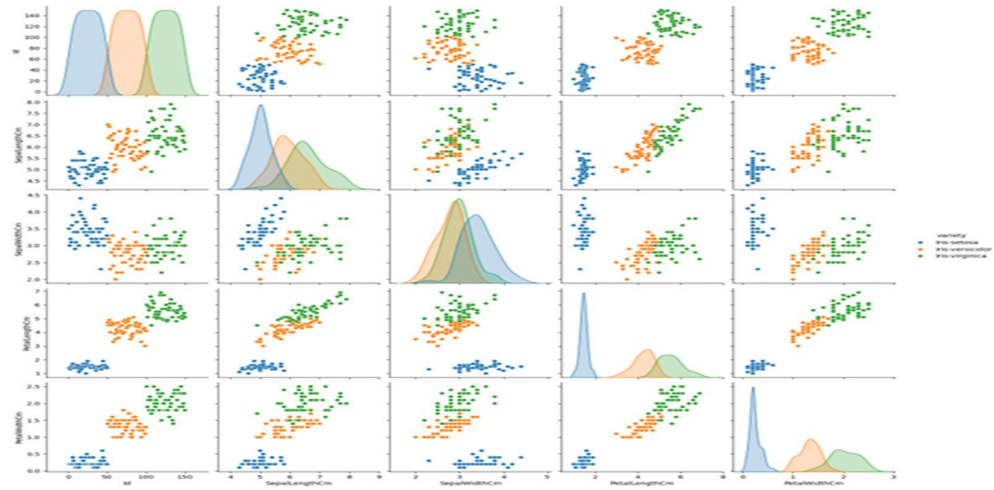
```
<Axes: xlabel='SepalLengthCm', ylabel='SepalWidthCm'>
```

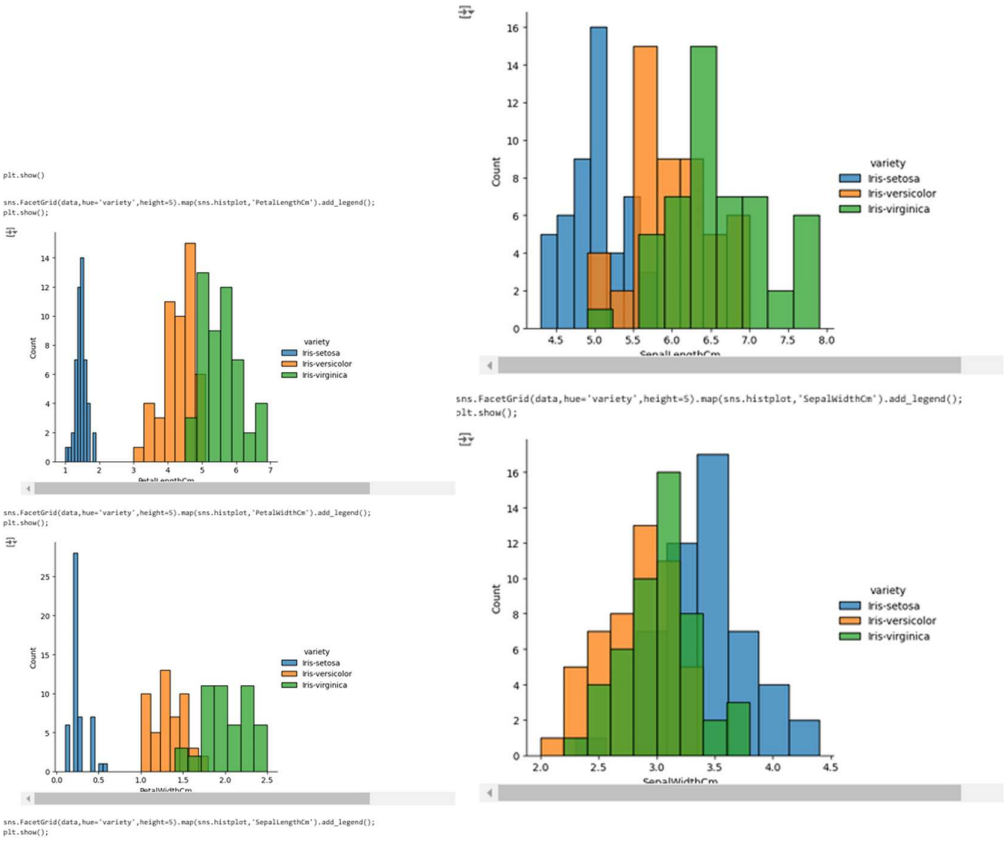



```
<Axes: xlabel='PetalLengthCm', ylabel='PetalWidthCm'>
```




```
sns.pairplot(data,hue='variety',height=3);
```





RESULT: Thus, the experiment to understand matplotlib library use cases in data science through visualization is completed.